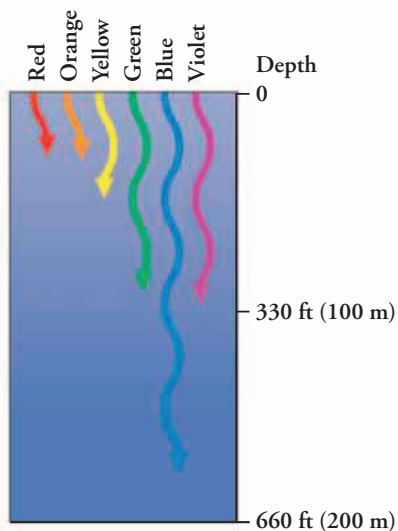




DEEP BLUE

Sunlight is made up of all the colors of the rainbow. Just beneath the waves, all these colors combine to form white light, but as the light penetrates deeper into the ocean, some of the colors are filtered out. Red, orange, and yellow are first to go, leaving only green, blue, and violet. Eventually, only blue light is left, but there is still enough of this to support plantlike life down to an average depth of 660 ft (200 m).

VITAL LIGHT

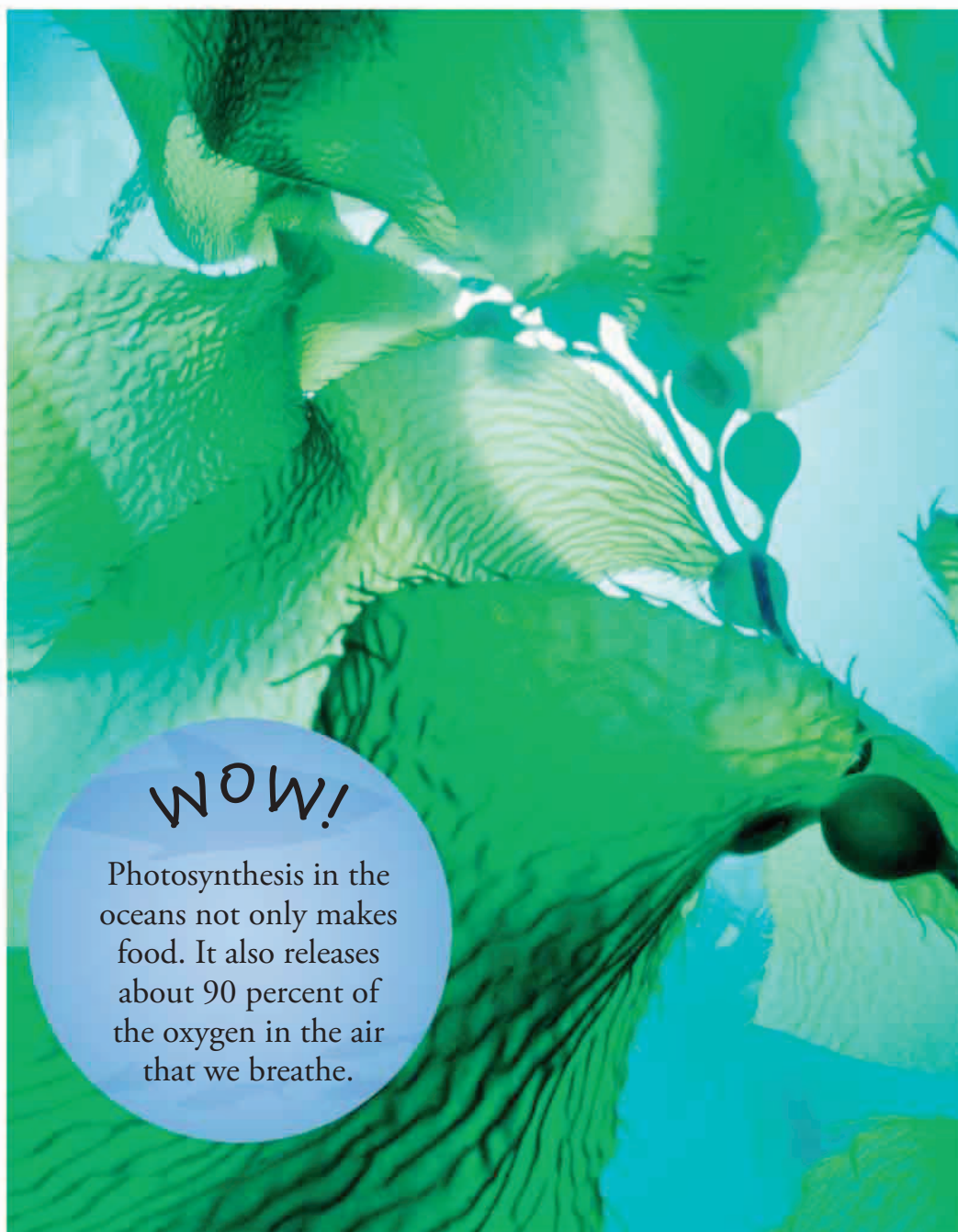
The plantlike organisms that live in the sunlit zone use the energy of light to make sugar compounds, which they turn into living tissue that can be eaten by animals. Most of these organisms are microscopic algae and special forms of bacteria that drift as phytoplankton in the open sea. But they also include the much bigger algae that we call seaweeds. Most of these seaweeds and the seagrass live attached to the seabed in shallow coastal water.

► GREEN GLOW

The light glowing through these seaweed fronds provides the essential energy that the seaweed needs to grow.

Sunlit zone

Most of the animals in the oceans live in the sunlit zone near the surface. They live here because, ultimately, nearly all animal life depends on the food made by plantlike seaweeds and tiny drifting organisms called phytoplankton. Just like land plants, these organisms cannot survive without sunlight. This forces them to live in the top 660 ft (200 m) of the ocean, where the light level is high enough for them to grow and multiply.



NOW!

Photosynthesis in the oceans not only makes food. It also releases about 90 percent of the oxygen in the air that we breathe.